Lesson 2.4: Cyclic Groups

Learning Outcomes: At the end of the lesson, the students are able to:

1. identify which groups are cyclic or not;
2. establish properties of cyclic groups;
3. determine the isomorphism classes of cyclic groups.
4. explore the subgroup structure of cyclic groups;

What is a cyclic group?

In Lesson 2.2, we have studied cyclic subgroups of G, which are subgroups of the form $\langle a \rangle$, where $a \in G$. In most cases, $\langle a \rangle$ is smaller than G itself. However, there are instances when $G = \langle a \rangle$. In this lesson, we shall study groups which can be generated by a single element.

Definition 2.4.1 (Cyclic Group)
A group G is **cyclic** if there exists $a \in G$ such that $G = \langle a \rangle$. In this case, the element a is called a **generator** of G.

In other words, a group G is cyclic if and only if it has a generator a. Consequently, every element of G may be expressed as a^k for some $k \in \mathbb{Z}$.

Example 2.4.2 The additive group $\mathbb{Z}$ is cyclic since $\mathbb{Z} = \langle 1 \rangle$. Also, $\mathbb{Z} = \langle -1 \rangle$. The generators of $\mathbb{Z}$ are 1 and -1.

Example 2.4.3 The group $\mathbb{Z}_6$ is cyclic. In Example 2.2.20, we have listed down the cyclic subgroups of $\mathbb{Z}_6$ and we have found out that

$$\mathbb{Z}_6 = \langle 1 \rangle = \langle 5 \rangle.$$

Thus, the elements 1 and 5 are generators of $\mathbb{Z}_6$.

In general, for any positive integer n, the group $\mathbb{Z}_n$ is cyclic as the element 1 is an automatic generator.

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Example 2.4.4 The Klein-4 group V_4 is not cyclic. This is because every non-identity element of V_4 has order 2. Indeed,

- $\langle e \rangle = \{e\}$
- $\langle b \rangle = \{e, b\}$
- $\langle a \rangle = \{e, a\}$
- $\langle c \rangle = \{e, c\}$

Therefore, no element can be a generator for V_4.

One important consequence of a group being cyclic is the following result.

Theorem 2.4.5 Cyclic groups are abelian.

Proof. Let G be a cyclic group. Then there exists $a \in G$ such that $G = \langle a \rangle$. Let $x, y \in G$. Then we may write $x = a^m$ and $y = a^n$, for some $m, n \in \mathbb{Z}$. Thus, by the law of exponents,

$$xy = a^m a^n = a^{m+n} = a^{n+m} = a^n a^m = yx.$$

Therefore, G is abelian. □

It is important to note that the proof of Theorem 2.4.5 is reliant on the fact that addition of integers itself is commutative.

Remark 2.4.6 If a group G is non-abelian, then G is not cyclic.

If our aim is to characterize all cyclic groups, then the non-abelian groups are automatically disqualified. However, there are abelian groups which are not cyclic. Then Klein-4 group V_4 is abelian, but not cyclic. If we are going to draw a map of all groups, then we can represent it as in Figure 8

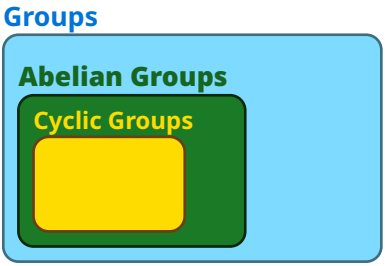


Figure 8: Map of All Groups

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Take note that being cyclic is a group structural property. Hence, if one group is cyclic and another one is not, then they cannot be isomorphic.

We see that the group $(\mathbb{Z}, +)$ is a cyclic group and that the set $\mathbb{Z}$ is countably infinite. Also, the set $\mathbb{Q}$ is countably infinite. We ask: is $(\mathbb{Z}, +)$ isomorphic to $(\mathbb{Q}, +)$?

The next example shows that $(\mathbb{Z}, +) \ncong (\mathbb{Q}, +)$.

Example 2.4.7 The group $(\mathbb{Q}, +)$ is not cyclic.

Proof. We argue indirectly. Suppose that $(\mathbb{Q}, +)$ is cyclic. Then there exists $r \in \mathbb{Q}$ such that $\mathbb{Q} = \langle r \rangle$. Write $r = \frac{p}{q}$, where $p \in \mathbb{Z}$ and $q \in \mathbb{Z}^+$. Assume, without loss of generality, that $\gcd(p, q) = 1$ so that the fraction $\frac{p}{q}$ is in simplest form.

Consider the rational number $s \coloneqq \frac{1}{q+1}$. Since $r = \frac{p}{q}$ is a generator, then there exists $n \in \mathbb{Z}$ such that $s = nr$. This means that

$$\frac{1}{q+1} = \frac{np}{q} \implies q = (np)(q+1).$$

However, the last equation implies that $q + 1$ is a divisor of q, which is not possible because q is a positive integer. This contradiction allows us to conclude that $(\mathbb{Q}, +)$ must not be cyclic. □

The next result shows that there aren't many "different" examples of cyclic groups, which is a direct application of Theorem 2.2.22.

Theorem 2.4.8 (Classification of Cyclic Groups) Let G be a cyclic group generated by a.

1. If a has an infinite order, then $G \cong \mathbb{Z}$.
2. If a has a finite order n, then $G \cong \mathbb{Z}_n$.

Consequently, two cyclic groups are isomorphic if and only if they have the same order.

Proof. Let $G = \langle a \rangle$.

1. If a has an infinite order, by Theorem 2.2.22, G is infinite and

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$$G = \{e, a, a^{-1}, a^2, a^{-2}, \dots\}.$$

Define $f : G \longrightarrow \mathbb{Z}$ as $f(a^n) \coloneqq n$, for all $n \in \mathbb{Z}$. The student now verifies that f is a group isomorphism so that $G \cong \mathbb{Z}$.

2. If $|a|_G = n$, by Theorem 2.2.22, we have $|G| = n$ and

$$G = \{e, a, a^2, \dots, a^{n-1}\}.$$

The function $g : G \longrightarrow \mathbb{Z}_n$ given by $g(a^k) = k$, for $k \in \{0, 1, \dots, n-1\}$ gives the necessary isomorphism, as the student will verify, so that $G \cong \mathbb{Z}_n$.□

Note 2.4.9 The intuition behind the isomorphisms f and g in the proof of Theorem 2.4.8 is that if we ignore the generator a and only consider the corresponding exponents, multiplication in G is merely just addition in $\mathbb{Z}$ or $\mathbb{Z}_n$, which is supported by the law of exponents.

Subgroup Structure of Cyclic Groups

We now know precisely which groups are cyclic and which are not. For a given cyclic group, we now characterize its subgroups. As we shall see in this section, the subgroup structure for cyclic groups is quite simple. We begin with the following result.

Theorem 2.4.10 Subgroups of cyclic groups are cyclic. In other words, the only subgroups of a cyclic group are its cyclic subgroups.

Proof. The theorem trivially holds when $G = \{e\}$ is a group of order 1 since the only subgroup in this case is $\{e\} = \langle e \rangle$. For the rest of the proof, assume that $|G| > 1$.

Assume that $G = \langle a \rangle$, for some $e \neq a \in G$. Let $\{e\} \neq H \leq G$. We claim that H is a cyclic subgroup by showing that $H = \langle c \rangle$, for some $c \in H$. To do this, we observe that every element of G is an integer power of a, hence, so are the elements of H since $H \subseteq G$. Also, since H is closed under inverses, there is at least one positive power of a that belongs to H.

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Let m be the smallest positive integer such that $a^m \in H$. We show that $c = a^m$ is a generator for H. To do this, let $h \in H$. Then $h = a^n$ for some $n \in \mathbb{Z}$. We apply Theorem 1.2.1 (division algorithm) to write $n = mq + r$, for some $q, r \in \mathbb{Z}$ with $0 \leq r < m$. In this way, we have

$$h = a^n = a^{mq+r} = (a^m)^q a^r,$$

from which it follows that $a^r = (a^m)^{-q} h \in H$ since $a^m, h \in H$, and the fact that H is a subgroup. Since $0 \leq r < m$, it follows that $r = 0$ by the definition of m. Thus,

$$h = (a^m)^q = c^q.$$

This proves that every element of H is an integer power of c, so $H = \langle c \rangle$.

Therefore, H is cyclic. □

Theorem 2.4.10 allows us to enumerate all the subgroups of a cyclic group G by simply enumerating all of its cyclic subgroups.

In the case when a cyclic group is infinite, it is enough to look at the group $(\mathbb{Z}, +)$ in view of Theorem 2.4.8. By considering the computations in Example 2.2.19, we have the following as a consequence.

Corollary 2.4.10.1 The only subgroups of $\mathbb{Z}$ are $\{0\}$ and the subgroups of the form $n\mathbb{Z}$, where n is a positive integer.

Another consequence of Theorem 2.4.10 when applied to the additive group $\mathbb{Z}$ is a result in number theory known as the *Bezout's identity*, which expresses the greatest common divisor of two numbers as a linear combination of the numbers. In a number theory course, this result is often proven using the *Euclidean algorithm*, which is a repeated application of division algorithm to obtain the greatest common divisor, but we shall prove it here using Theorem 2.4.10.

Corollary 2.4.10.2 (Bezout's Identity)
Let $n, m \in \mathbb{Z}$ and let $d = \gcd(n, m)$. Then there exist $r, s \in \mathbb{Z}$ such that

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$$d = nr + ms.$$

Proof. Let $n, m \in \mathbb{Z}$ and let $d = \gcd(n, m)$. Define

$$H = \{nr + ms \mid r, s \in \mathbb{Z}\}.$$

We show that $H \leq \mathbb{Z}$. First of all, $0 = n(0) + m(0) \in H$ so H is nonempty. Let $h_1, h_2 \in H$. Write

$$h_1 = nr_1 + ms_1 \quad \text{and} \quad h_2 = nr_2 + ms_2,$$

where $r_1, r_2, s_1, s_2 \in \mathbb{Z}$. Then

$$\begin{aligned} h_1 + (-h_2) &= nr_1 + ms_1 - (nr_2 + ms_2) \\ &= n(r_1 - r_2) + m(s_1 - s_2). \end{aligned}$$

Since $r_1 - r_2 \in \mathbb{Z}$ and $s_1 - s_2 \in \mathbb{Z}$, it follows that $h_1 + (-h_2) \in H$. By Theorem 2.2.7, it is true that $H \leq \mathbb{Z}$.

We now invoke Theorem 2.4.10 and conclude that

$$H = \{nr + ms \mid r, s \in \mathbb{Z}\} = \langle d' \rangle = d'\mathbb{Z},$$

for some integer $d' > 0$. We claim that $d' = d$.

If we set $r = 1$ and $s = 0$, then $nr + ms = n \in H = \langle d' \rangle$. Thus, $d' \mid n$. Similarly, if we set $r = 0$ and $s = 1$, then $nr + ms = m \in H = \langle d' \rangle$. Thus, $d' \mid m$. Hence, d' is a common divisor of n and m. Since $d = \gcd(n, m)$, it follows that $d' \leq d$. Now, note that $d' \in H$. Thus, there exist $r_0, s_0 \in \mathbb{Z}$ such that

$$d' = nr_0 + ms_0.$$

Since $d = \gcd(n, m)$, it follows that $d \mid nr_0$ and $d \mid ms_0$. Consequently, $d \mid d'$. The fact that $d, d' > 0$ implies that $d \leq d'$. Therefore, $d = d'$.

Finally, the fact that $d \in \langle d \rangle = \{nr + ms \mid r, s \in \mathbb{Z}\}$ implies the conclusion of the theorem. □

Note 2.4.11 If $n, m \in \mathbb{Z}$, we also write

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$$H = \{nr + ms \mid r, s \in \mathbb{Z}\} =: n\mathbb{Z} + m\mathbb{Z}.$$

Consequently,

$$n\mathbb{Z} + m\mathbb{Z} = d\mathbb{Z} \iff d = \gcd(n, m).$$

Using Note 2.2.14 and Note 2.4.11, the relationship among the subgroups $n\mathbb{Z}$, $m\mathbb{Z}$, $n\mathbb{Z} + m\mathbb{Z}$ and $n\mathbb{Z} \cap m\mathbb{Z}$ may be visualized as a lattice diagram shown in Figure 9.

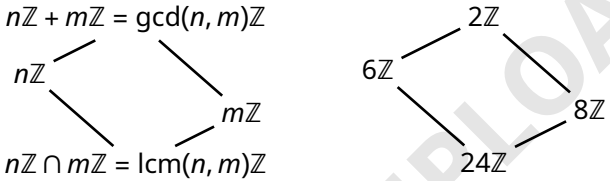


Figure 9: The Lattice of Subgroups $n\mathbb{Z}$ and $m\mathbb{Z}$

Subgroups of Finite Cyclic Groups

The finite cyclic group case is a more complicated situation and requires more elaborate computations. To get a glimpse of this idea, consider the following example.

Example 2.4.12 List down the distinct subgroups of $\mathbb{Z}_{12}$ and sketch its lattice diagram of subgroups.

Solution. Since $\mathbb{Z}_{12}$ is a cyclic group, by Theorem 2.4.10, it is enough to enumerate all of its cyclic subgroups:

- $\langle 0 \rangle = \{0\}$
- $\langle 1 \rangle = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 0\} = \mathbb{Z}_{12}$ (1 is a generator.)
- $\langle 2 \rangle = \{2, 4, 6, 8, 10, 0\}$
- $\langle 3 \rangle = \{3, 6, 9, 0\}$
- $\langle 4 \rangle = \{4, 8, 0\}$
- $\langle 5 \rangle = \{5, 10, 3, 8, 1, 6, 11, 4, 9, 2, 7, 0\} = \mathbb{Z}_{12}$ (5 is a generator.)

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- $\langle 6 \rangle = \{6, 0\}$
- $\langle 7 \rangle = \{7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, 0\} = \mathbb{Z}_{12}$ (7 is a generator.)
- $\langle 8 \rangle = \{8, 4, 0\}$
- $\langle 9 \rangle = \{9, 6, 3, 0\}$
- $\langle 10 \rangle = \{10, 8, 6, 4, 2, 0\}$
- $\langle 11 \rangle = \{11, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1, 0\} = \mathbb{Z}_{12}$ (11 is a generator.)

In summary, we see that $\mathbb{Z}_{12}$ only has 6 distinct subgroups namely,

1. $\mathbb{Z}_{12} = \langle 1 \rangle = \langle 5 \rangle = \langle 7 \rangle = \langle 11 \rangle$
2. $\{0, 2, 4, 6, 8, 10\} = \langle 2 \rangle = \langle 10 \rangle$
3. $\{0, 3, 6, 9\} = \langle 3 \rangle = \langle 9 \rangle$
4. $\{0, 4, 8\} = \langle 4 \rangle = \langle 8 \rangle$
5. $\{0, 6\} = \langle 6 \rangle$
6. $\{0\} = \langle 0 \rangle$

By looking at how these subgroups are related in terms of set inclusions, we can sketch the lattice diagram of subgroups for $\mathbb{Z}_{12}$ as in Figure 10.

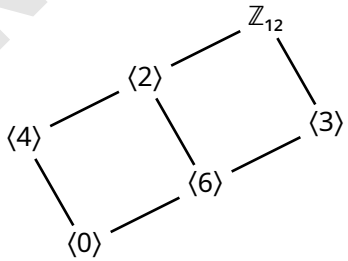


Figure 10: Lattice Diagram of Subgroups for $\mathbb{Z}_{12}$

The method of listing subgroups presented in Example 2.4.12 is exhaustive and requires you to check the cyclic subgroups generated by each element of $\mathbb{Z}_{12}$. We can only imagine how painfully long the process is for the group $\mathbb{Z}_n$, when n is *large*.

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In this section, we shall answer the following questions for a finite cyclic group G:

1. What is the order of a given element of G?
2. When is an element of G a generator? How many generators are there?
3. How many subgroups does G have?

Recall from Theorem 2.2.22 that if G is a finite cyclic group of order n that is generated by a, then a is an element of order n and

$$G = \{e, a, a^2, \ldots, a^{n-1}\}.$$

Theorem 2.4.13 Let $G = \langle a \rangle$ be a cyclic group of order n. The following hold:

1. If $k \in \mathbb{Z}$ and $\gcd(n, k) = d$, then $\langle a^k \rangle = \langle a^d \rangle$. Consequently,
$$|a^k|_G = |a^d|_G.$$
2. If $k \in \mathbb{Z}$ and $\gcd(n, k) = d$, then
$$|a^k|_G = \frac{n}{d} = \frac{n}{\gcd(n, k)}.$$

Proof. Let $k \in \mathbb{Z}$ and $d = \gcd(n, k)$. Since $d \mid k$, we may write $k = d\ell$, for some $\ell \in \mathbb{Z}$, so that $a^k = (a^d)^\ell$. This means that every power of a^k is also a power of a^d proving $\langle a^k \rangle \subseteq \langle a^d \rangle$. To prove the reverse inclusion, we use Corollary 2.4.10.2 to write $d = nr + ks$, for some $r, s \in \mathbb{Z}$. In this case, we have

$$a^d = a^{nr+ks} = (a^n)^r (a^k)^s = e\,(a^k)^s = (a^k)^s.$$

Thus, every power of a^d is also a power of a^k so $\langle a^k \rangle \subseteq \langle a^d \rangle$. Therefore, $\langle a^k \rangle = \langle a^d \rangle$.

To prove the second statement, it is enough to show that $|a^d|_G = n/d$, because $|a^k|_G = |a^d|_G$. Note that

$$(a^d)^{n/d} = a^n = e,$$

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implying that $|a^d|_G \leq \frac{n}{d}$. We shall argue that the alternative $|a^d|_G < \frac{n}{d}$ leads to a contradiction. Suppose that $|a^d|_G = m < \frac{n}{d}$ holds. Then $dm < n$. Since $|a|_G = n$, it follows by definition that $(a^d)^m = a^{dm} \neq e$. This contradicts the assumption that $|a^d|_G = m$. Therefore,

$$|a^k|_G = |a^d|_G = \frac{n}{d} = \frac{n}{\gcd(n,k)}.$$ □

Corollary 2.4.13.1 If G is a finite cyclic group and $H \leq G$, then $|H|$ divides $|G|$.

Proof. Assume $G = \langle a \rangle$, where $|a|_G = n$. Let $H \leq G$. Theorem 2.4.10 guarantees that $H = \langle a^k \rangle$, for some $k \in \mathbb{Z}$. By Theorem 2.4.13,

$$|H| = |a^k| = \frac{n}{d}, \quad \text{where } d = \gcd(n,k).$$

If we let $|H| = m$, then $m = \frac{n}{d}$ implying that $md = n$, so $m \mid n$ as we wish. □

Corollary 2.4.13.2 Let $G = \langle a \rangle$ be a finite cyclic group of order n. Then

1. $\langle a^k \rangle = \langle a^\ell \rangle$ if and only if $\gcd(n,k) = \gcd(n,l)$.
2. a^k is a generator if and only if $\gcd(n,k) = 1$.

Proof. To prove statement (1), we directly apply statement (1) of Theorem 2.4.13. Since $\langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle$ and $\langle a^\ell \rangle = \langle a^{\gcd(n,\ell)} \rangle$, we only need to prove that

$$\langle a^{\gcd(n,k)} \rangle = \langle a^{\gcd(n,\ell)} \rangle \iff \gcd(n,k) = \gcd(n,l).$$

The backward implication is certainly true. Assume that $\langle a^{\gcd(n,k)} \rangle = \langle a^{\gcd(n,\ell)} \rangle$. By statement (2) of Theorem 2.4.13,

$$|a^{\gcd(n,k)}|_G = |a^{\gcd(n,\ell)}|_G \implies \frac{n}{\gcd(n,k)} = \frac{n}{\gcd(n,\ell)} \implies \gcd(n,k) = \gcd(n,\ell).$$

To prove statement (2), we apply statement (2) of Theorem 2.4.13 as follows:

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$$G = \langle a^k \rangle \iff |a^k|_G = n \iff \frac{n}{\gcd(n,k)} = n \iff \gcd(n,k) = 1. \qquad \square$$

When these results apply to the cyclic group $\mathbb{Z}_n$, take note that we may take $a = 1$ so that

$$\mathbb{Z}_n = \langle 1 \rangle = \{0, 1, 2, \ldots, n-1\}.$$

Note that every power a^k corresponds to ka modulo n. We state the following as a remark.

Remark 2.4.14 In the group $\mathbb{Z}_n$,

1. The order of $k \in \mathbb{Z}_n$ is $\frac{n}{\gcd(n,k)}$.
2. $\langle k \rangle = \langle \ell \rangle$ if and only if $\gcd(n,k) = \gcd(n,\ell)$.
3. $k \in \mathbb{Z}_n$ is a generator if and only if $\gcd(n,k) = 1$.

The student is asked to re-examine Example 2.4.12 and verify the statements in Remark 2.4.14.

Let $U(n)$ denote the set of all generators for the group $\mathbb{Z}_n$. Statement (3) of Remark 2.4.14 implies

$$U(n) = \{k \mid 0 < k < n \text{ and } \gcd(n,k) = 1\}$$

is the set of all positive integers less than n which are relatively prime to n.

Example 2.4.15 Determine all the generators of the group $\mathbb{Z}_{20}$.

Solution. The generators of $\mathbb{Z}_{20}$ are the elements of $U(20)$, which consists of all positive integers less than 20, which are relatively prime with 20. Checking each integer one-by-one, we see that

$$U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}.$$

As a matter of generalization, we determine the number of elements in $U(n)$, for an arbitrary n. The cardinality of $U(n)$ is equal to the number of positive integers less than n, which are relatively prime to n. In number theory, this number is denoted by $\varphi(n)$, where φ is the function known as the *Euler totient function* or simply the *Euler-Phi function*. Thus, $|U(n)| = \varphi(n)$.

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For small values of n, the values of $\varphi(n)$ are shown in Table 12 and the plot for $n \leq 100$ is shown in Figure 11.

n	2	3	4	5	6	7	8	9	10
$\varphi(n)$	1	2	2	4	2	6	4	6	4

Table 12: Values of $\varphi(n)$, for $n \leq 10$

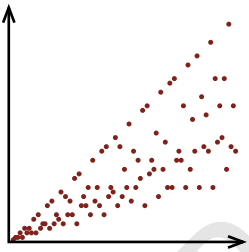


Figure 11: Plot of $\varphi(n)$, for $n \leq 100$

In order for us to efficiently calculate specific values of $\varphi(n)$, we shall use the following properties of $\varphi(n)$, which are typically covered in an elementary course in number theory. These properties will be stated without proofs.

Note 2.4.16 For any positive integer $n > 1$,

1. $\varphi(p) = p - 1$, if p is a prime.
2. $\varphi(p^k) = p^k - p^{k-1}$, if p is a prime and $k > 0$ is an integer.
3. $\varphi(mn) = \varphi(m)\varphi(n)$, if $\gcd(m, n) = 1$.

Using these properties, we may utilize the prime factorization of an integer $n > 1$ to calculate $\varphi(n)$. If $n = p_1^{k_1} p_2^{k_2} \ldots p_m^{k_m}$ is the prime factorization of n, where p_i's are distinct primes, then we may calculate $\varphi(n)$ as follows:

$$\begin{aligned}\varphi(n) &= \varphi\left(p_1^{k_1}\right)\varphi\left(p_2^{k_2}\right)\ldots\varphi\left(p_m^{k_m}\right)\\ &= \left(p_1^{k_1} - p_1^{k_2-1}\right)\left(p_2^{k_1} - p_2^{k_2-1}\right)\ldots\left(p_m^{k_m} - p_m^{k_m-1}\right)\\ &= \prod_{i=1}^{m}\left(p_i^{k_i} - p_i^{k_i-1}\right).\end{aligned}$$

As a matter of illustration, consider the case when $n = 20$. Since $20 = 2^2 \cdot 5$, applying the formula above yields

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$$\begin{aligned}\varphi(20) &= \varphi(2^2)\varphi(5)\\ &= (2^2 - 2^1)(5-1)\\ &= 2 \cdot 4\\ &= 8.\end{aligned}$$

Notice that this number is consistent with our manual listing of the elements of

$$U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}.$$

We note that the number of generators for $\mathbb{Z}_n$ depends on the prime factorization of n. Although, prime factorizations are hard to achieve for big values of n, this method of counting is a lot more effective than manually checking all the positive integers less than n which are relatively prime to n.

We now attempt to completely characterize the subgroups of a finite cyclic group. Corollary 2.4.13.1 guarantees that if H is a subgroup of a finite cyclic group G, then the order of H is a divisor of the order of G. In fact, we can make this result even stronger.

Theorem 2.4.17 (Subgroups of a Finite Cyclic Group) Let G be a finite cyclic group of order n. For each positive divisor d of n, there exists a unique subgroup H of G whose order is d. Consequently, the number of subgroups of G is equal to the number of positive divisors of n.

Proof. Let G be a finite cyclic group of order n, generated by an element a. Let d be a positive divisor of n. For the proof of existence, take $H = \langle a^{n/d} \rangle$. Then H is a subgroup of G and by Theorem 2.4.13,

$$|H| = \left|a^{n/d}\right|_G = \frac{n}{\gcd(n, n/d)} = \frac{n}{n/d} = d.$$

For the proof of uniqueness, let $H' = \langle a^k \rangle$ be a subgroup of G with d elements. Theorem 2.4.13 guarantees that $\frac{n}{\gcd(n,k)} = d$ implying that $\frac{n}{d} = \gcd(n, k)$. Consequently,

$$H' = \langle a^k \rangle = \langle a^{\gcd(n,k)} \rangle = \langle a^{n/d} \rangle = H. \qquad \square$$

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Remark 2.4.18 In the group $\mathbb{Z}_n$, for each divisor d of n, the unique subgroup of order d is $\langle n/d \rangle$.

Example 2.4.19 Enumerate the distinct subgroups of $\mathbb{Z}_{30}$.

Solution. Each subgroup of $\mathbb{Z}_{30}$ corresponds to a positive divisor of 30, so we first list down the positve divisors of 30, namely 1, 2, 3, 5, 6, 10, 15, 30.

d	n/d	subgroup
1	30	$\langle 0 \rangle = \{0\}$ Note: $30 \equiv 0 \pmod{30}$
2	15	$\langle 15 \rangle = \{0, 15\}$
3	10	$\langle 10 \rangle = \{0, 10, 20\}$
5	6	$\langle 6 \rangle = \{0, 6, 12, 18, 24\}$
6	5	$\langle 5 \rangle = \{0, 5, 10, 15, 20, 25\}$
10	3	$\langle 3 \rangle = \{0, 3, 6, 9, 12, 15, 18, 21, 24, 27\}$
15	2	$\langle 2 \rangle = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28\}$
30	1	$\langle 1 \rangle = \mathbb{Z}_{30}$

The lattice diagram of subgroups for $\mathbb{Z}_{30}$ is shown in

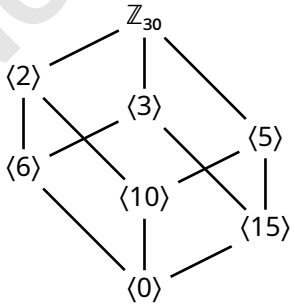


Figure 12: Lattice Diagram of Subgroups for $\mathbb{Z}_{30}$

The Group of Units

We have already seen the group and subgroup structure of the cyclic group $\mathbb{Z}_n$ under addition modulo n. Aside from addition modulo n, in Lesson 1.2, we have also learned about multiplication modulo n, which is also an

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associative binary operation on $\mathbb{Z}_n$. Furthermore, the element $1 \in \mathbb{Z}_n$ is the identity element under this operation. What stops us from claiming that $\mathbb{Z}_n$ is a group under multiplication modulo n is the fact that not every element in invertible.

For example, if we consider $\mathbb{Z}_6$ under multiplication modulo 6 and its Cayley table in Table 5 we see that only the elements of 1 and 5 are invertible and that $1^{-1} = 1$ and $5^{-1} = 5$ with respective to multiplication modulo 6. We are now about to solve the general problem of determining a necessary and sufficient condition under which an element of $\mathbb{Z}_n$ is invertible under multiplication modulo n.

Theorem 2.4.20 An element $m \in \mathbb{Z}_n$ is invertible under multiplication modulo n if and only if $m \in U(n)$.

Proof. Assume that $m \in \mathbb{Z}_n$ is invertible under multiplication modulo n. Then there exists $r \in \mathbb{Z}_n$ such that $mr \equiv 1 \pmod{n}$. This means $n \mid mr - 1$. Thus, there exists $k \in \mathbb{Z}$ such that $nk = mr - 1$ or that

$$mr - nk = 1.$$

Let $d = \gcd(m, n)$. Then $d \mid mn$ and $d \mid nk$ implying that $d \mid mr - nk$. Thus, $d \mid 1$. Therefore, $g(m, n) = d = 1$ so $m \in U(n)$.

Conversely, assume that $m \in U(n)$. Then $\gcd(m, n) = 1$. By Corollary 2.4.10.2, there exists $r, s \in \mathbb{Z}$ such that $mr + ns = 1$. Since $ns \equiv 0 \pmod{n}$, it follows that

$$mr \equiv mr + ns \pmod{n} \equiv 1 \pmod{n}.$$

Therefore, m is invertible under multiplication modulo n. □

Corollary 2.4.20.1 For $n > 1$, the set $U(n)$ is a group under multiplication modulo n.

Example 2.4.21 Determine if the group $U(20)$ is cyclic.

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Solution. Recall that $U(20) = \{1, 3, 7, 9, 11, 13, 17, 19\}$. Under multiplication modulo 20, we compute powers modulo 20 and found the following cyclic subgroups:

- $\langle 1 \rangle = \{1\}$
- $\langle 3 \rangle = \{1, 3, 9, 7\}$
- $\langle 7 \rangle = \{1, 7, 9, 3\}$
- $\langle 9 \rangle = \{1, 9\}$
- $\langle 11 \rangle = \{1, 11\}$
- $\langle 13 \rangle = \{1, 13, 9, 17\}$
- $\langle 17 \rangle = \{1, 17, 9, 13\}$
- $\langle 19 \rangle = \{1, 19\}$

As we can see, no element has an order 8. Therefore, $U(20)$ is not cyclic.

Can you find other subgroups of $U(20)$?

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